EQUITY IN AUTOIMMUNITY:

MAPPING OVERLAPPING GENETIC VARIANTS IN UNDERSERVED POPULATIONS.

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FOCUS TOPICS

TYPE 1 DIABETES

THE BLACK COMMUNITY

ADOLESCENTS

REPRESENTATION OF ALL 3 COMBINED

PROPOSED RESEARCH QUESTION:

What shared variants/loci exist between severe early-onset allergic phenotypes and T1D, and are these loci underrepresented in Black populations in the national institute of health public archive (clinvar)?

THE DATA

ClinVar, National Library of Medicine (NCBI)
Varaint_Summary data

GWAS Catalog, National Human Genome Research Institute (NHGRI) & EMBL-EBI
Association data
Ancestry data

VARIANT_DATA


```
Rows: 8,887,816
Columns: 43
```

```
sum(is.na(Variant_data))
```

```
## [1] 0
```

```
character    numeric
       33         10
```

Category	Primary Function	Key Fields (Representative)
Core Identifiers	What is the variant & gene?	AlleleID, Name, GeneSymbol
Clinical Significance	What does it mean for health?	ClinicalSignificance, ReviewStatus, LastEvaluated
Phenotype & Origin	What disease/condition?	PhenotypeList, Origin
Genomic Location	Where is it on the genome?	Chromosome, Start, Stop, Assembly
Cross-References	Connectivity to other DBs	RS#(dbSNP), RCVaccession, VariationID
Advanced/Technical	VCF formatting & specialized	PositionVCF, SomaticClinicalImpact, Oncogenicity

VARIANT_DATA

Type Diabetes

```
Rows: 3,862
Columns: 43
```

Severe Allergies

```
Rows: 286
Columns: 43
```

```
t1d_variants <- Variant_data %>%
  filter(str_detect(PhenotypeList, regex("type 1 diabetes|T1D|IDDM|insulin-dependent diabetes|diabetes mellitus|type 1 diabetes mellitus
20|monogenic autoimmune diabetes|autoimmune diabetes|neonatal diabetes mellitus|immune related diabetes|MODY|maturity-onset diabetes of the
young|maturity onset diabetes mellitus in young", ignore_case = TRUE))) %>% filter(str_detect(GeneSymbol,
regex("HLA-DRB1|HLA-DQA1|HLA-DQB1|INS|CTLA4|PTPN22|HNF1A|GCK|HNF4A", ignore_case = TRUE)))
```

```
anaphylaxis <- Variant_data %>%
  filter(str_detect(PhenotypeList, regex("mastocytosis|Hereditary alpha-tryptasemia|HαT|Idiopathic anaphylaxis|Mast cell activation syndrome",
ignore_case = TRUE))) %>% filter(str_detect(GeneSymbol, regex("KIT|TPSAB1|FCER1A|(IL4&IL3)", ignore_case = TRUE)))
```

Researched terms frequently used in conjunction with type 1 diabetes and anaphylaxis

Phenotypes:
type 1 diabetes
T1D
IDDM
insulin-dependent diabetes
diabetes mellitus
type 1 diabetes mellitus 20
monogenic autoimmune diabetes
autoimmune diabetes
neonatal diabetes mellitus
immune related diabetes
MODY
maturity-onset diabetes of the young
maturity onset diabetes mellitus in young

GeneSymbols:
HLA-DRB1
HLA-DQA1
HLA-DQB1
INS
CTLA4
PTPN22
HNF1A
GCK
HNF4A

Phenotypes:
Tymastocytosis
Hereditary
Alpha-tryptasemia
HaT
Idiopathic
anaphylaxis
Mast cell activation
syndrome

GeneSymbols:
KIT
TPSAB1
FCER1A
IL4&IL3

ACCESSION_DATA

```
character
       38
```

```
Rows: 1,061,816
Columns: 38
```

```
sum(is.na(gwas))
```

```
[1] 7693139
```

Category	Primary Function	Representative Fields
Study Context	Who did the research & where?	First Author, Journal, Date, Study Accession
Cohort Profile**	Who was studied?	Disease/Trait, Initial/Replication Sample Size
Genetic Location**	Where is the variant?	Chromosome, CHR_POS, Mapped_Gene, Context
Statistical Evidence**	Is the result reliable?	P-Value, OR or Beta, Risk Allele Frequency

ANCESTRY_DATA

```
character
       38
```

```
Rows: 39,980
Columns: 17
```

```
sum(is.na(gwas))
```

```
[1] 7693139
```

Category	Primary Function	Representative Fields
Study Identifiers	How to track/cite the research	Study Accession, PubMed ID, First Author
Study Design**	Which phase and purpose	Stage, Number of Cases/Controls
Population Data**	Who was included?	Broad Ancestry, Sample Description
Study Context	Where and when	Date, Country of Recruitment

ANCESTRY_DATA

```
ancestry_processed <- gwas_ancestor %>%
  mutate(has_african_ancestry = str_detect(
    `BROAD ANCESTRAL CATEGORY`,
    "African American or Afro-Caribbean|African unspecified"
  ))

# Now you can filter your main dataset for these studies
african_ancestry_studies <- ancestry_processed %>%
  filter(has_african_ancestry == TRUE)
```

```
Rows: 43,611
Columns: 18
```

```
Rows: 39,980
Columns: 17
```

```
gwas_ancestor <- gwas_ancestor%>% filter(`BROAD ANCESTRAL CATEGORY`== 'African American or Afro-Caribbean' | `BROAD ANCESTRAL CATEGORY`== 'African American' | `BROAD ANCESTRAL CATEGORY`== 'African unspecified'|`BROAD ANCESTRAL CATEGORY`== 'Afro-Caribbean' |`BROAD ANCESTRAL CATEGORY`== 'Sub-Saharan African'| `BROAD ANCESTRAL CATEGORY`== 'African unspecified, African American or Afro-Caribbean'| `BROAD ANCESTRAL CATEGORY`=='African American or Afro-Caribbean, NR'| `BROAD ANCESTRAL CATEGORY`== 'African American or Afro-Caribbean, African unspecified'|`BROAD ANCESTRAL CATEGORY`=='  African unspecified, NR'|`BROAD ANCESTRAL CATEGORY`=='NR, African unspecified'|`BROAD ANCESTRAL CATEGORY`== ' Sub-Saharan African, African American or Afro-Caribbean'|`BROAD ANCESTRAL CATEGORY`== 'Sub-Saharan African, African unspecified')
```

- Need a more efficient filtering approach to isolate studies with predominantly Black participant populations without relying on overly narrow or overly broad criteria

JOINED_GWAS

```
joined_gwas <- gwas_ancestor %>%
  inner_join(gwas)
joined_gwas
```

```
sum(is.na(gwas))
```

```
[1] 7693139
```

```
character
       52
```

```
Rows: 214,079
Columns: 52
```

```
#t1d filter
t1d_subset <- joined_gwas %>% filter(str_detect(`DISEASE/TRAIT`, "type 1 diabetes"))
#Severe allergy filter
allergy_subset <- joined_gwas %>% filter(str_detect(`DISEASE/TRAIT`, "mastocytosis|anaphylaxis|allergy"))
```

```
Rows: 7
Columns: 52
```

```
Rows: 19
Columns: 52
```

```
# Expanded the search
#t1d filter
t1d_subset <- joined_gwas %>% filter(str_detect(`DISEASE/TRAIT`, "type 1 diabetes|T1D|IDDM|insulin-dependent diabetes|diabetes mellitus|type 1 diabetes mellitus 20|monogenic autoimmune diabetes|autoimmune diabetes|neonatal diabetes mellitus|immune related diabetes|MODY|maturity-onset diabetes of the young|maturity onset diabetes mellitus in young"))
#severe allergy filter
allergy_subset <- joined_gwas %>% filter(str_detect(`DISEASE/TRAIT`, "mastocytosis|Hereditary alpha-tryptasemia|HαT|Idiopathic anaphylaxis|Mast cell activation syndrome|mastocytosis|anaphylaxis|allergy"))
```

```
Rows: 246
Columns: 52
```

```
Rows: 19
Columns: 52
```

JOINED_GWAS

A tibble: 0 × 103

0 rows | 1-8 of 103 columns

A tibble: 0 × 103

0 rows | 1-8 of 103 columns

```
# Find shared variants
shared_loci <- inner_join(t1d_subset, allergy_subset, by = "SNPS")
```

My current finding is there is no overlap in between the diabetes GWAS data and the allergy GWAS data.

Is this because there scientifically is no overlap?

Is this because of the intense filters I put on the demographics?

Is this because I didn't include all the indicators for either type 1 or anaphylaxis?

Is this a representation issue throughout the GWAS data?

Would this still be true if applied to the Variant data?

TIMELINE

This week

Adjust research question

Join GWAS and Variant data

Weeks 3-4

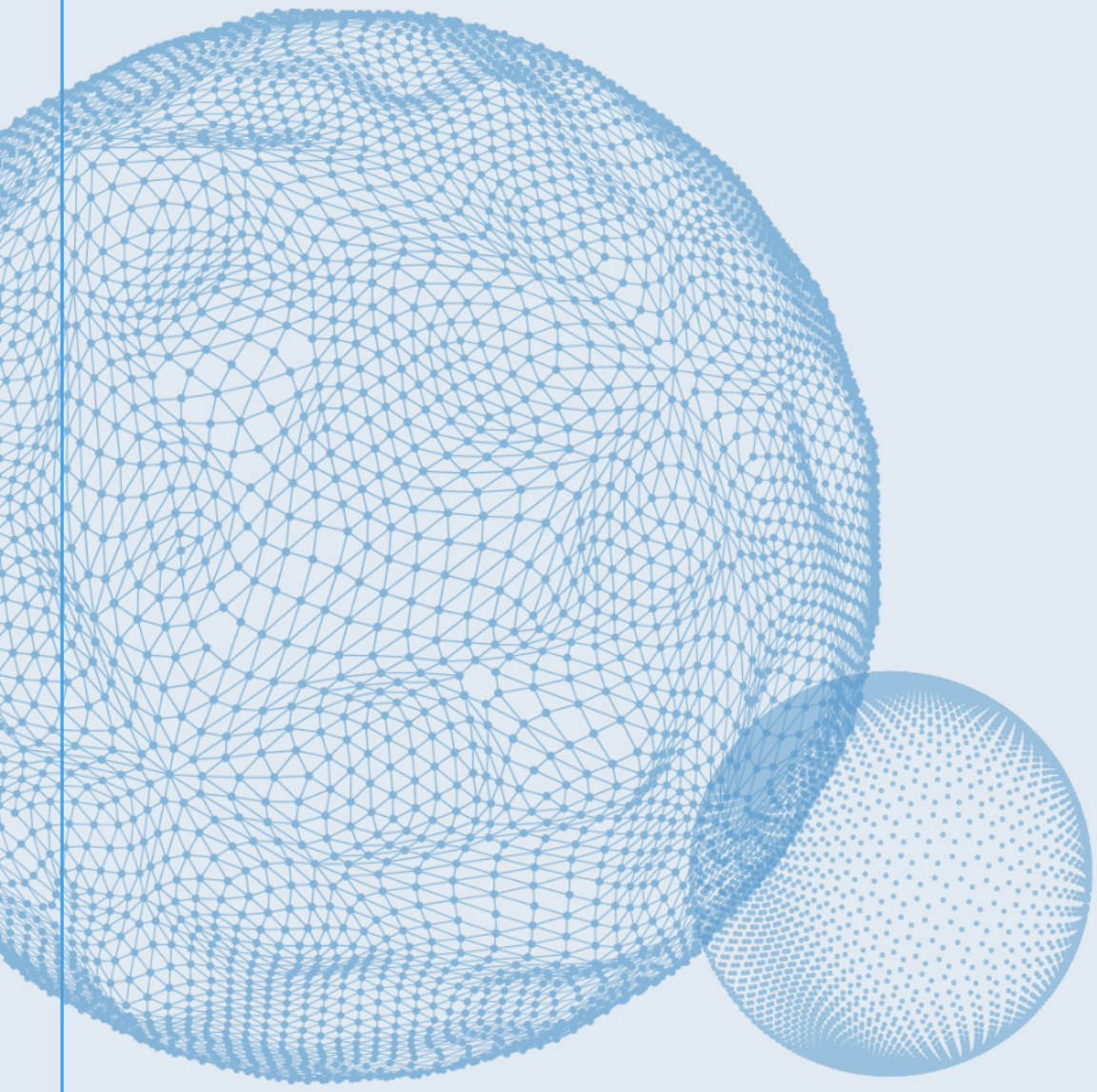
Leverage LLMs for data cleaning and standardizing

Weeks 5-7

Explore with Machine Learning Models for dimension reduction, clustering, or classification

Weeks 8-9

Finalize interpretation of findings and report



THANK YOU

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